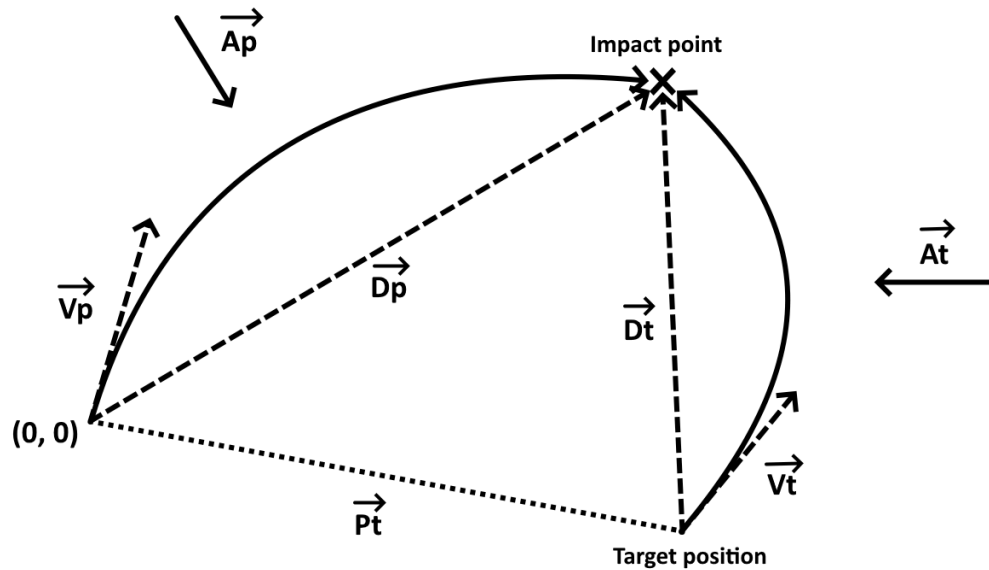


Mathematical report



Variables:

$v = |\vec{V}_p|$ – Projectile speed

$\hat{V}_p$ – Projectile direction vector

$\vec{V}_p = \hat{V}_p v$ – Projectile velocity vector

$\vec{A}_p$ – Projectile acceleration vector

$\vec{D}_p$ – Projectile displacement vector

$\vec{P}_t$ – Vector to relative target position

$\vec{V}_t$ – Target velocity vector

$\vec{A}_t$ – Target acceleration vector

$\vec{D}_t$ – Target displacement vector

t – Time to hit

$\vec{A} = \vec{A}_p - \vec{A}_t$ – Relative acceleration vector

General formulas:

$$D_p = V_p t + \frac{A_p t^2}{2} = t(V_p + \frac{A_p t}{2})$$

$$D_t = V_t t + \frac{A_t t^2}{2} = t(V_t + \frac{A_t t}{2})$$

$$D_p = P_t + D_t$$

$$V_p^2 = v^2$$

$$V_p t + \frac{A_p t^2}{2} = P_t + V_t t + \frac{A_t * t^2}{2}$$

$$\hat{V}_p v = V_p = \frac{P_t}{t} + V_t - \frac{A_t t}{2}$$

Finding time

Knowing speed

If we know projectile speed $v = |V_p|$ we can find time:

$$V_p^2 = v^2$$

$$V_p = \frac{P_t}{t} + V_t - \frac{A_t t}{2}$$

$$(\frac{P_t}{t} + V_t - \frac{A_t t}{2})^2 = v^2$$

$$(\frac{A_t t^2}{2} - V_t t - P_t)^2 - v^2 t^2 = 0$$

$$(\frac{A^2}{4})t^4 + (-AV_t)t^3 + (V_t^2 - AP_t - v^2)t^2 + (2V_t P_t)t + P_t^2 = 0$$

We need to solve this equation of 4th degree.

Knowing direction vector

$$\hat{V}_p vt = P_t + V_t t - \frac{At^2}{2}$$

$$v = \hat{V}_p \frac{P_t}{t} + \hat{V}_p V_t - \hat{V}_p \frac{At}{2}$$

$$R(t) = P_t + V_t t - \frac{At^2}{2}$$

Intersection is possible only if $R(t)$ lies along $\hat{V}_p$

$$R(t) - \hat{V}_p(\hat{V}_p \cdot R(t)) = 0$$

$$\left(\frac{A - (\hat{V}_p \cdot A)\hat{V}_p}{2}\right)t^2 - (V_t - (\hat{V}_p \cdot V_t)\hat{V}_p)t - (P_t - (\hat{V}_p \cdot P_t)\hat{V}_p) = 0$$

$$B = A - (\hat{V}_p \cdot A)\hat{V}_p$$

$$C = V_t - (\hat{V}_p \cdot V_t)\hat{V}_p$$

$$D = P_t - (\hat{V}_p \cdot P_t)\hat{V}_p$$

$$\frac{B}{2}t^2 - Ct - D = 0$$

$$\left(\frac{B^2}{4}\right)t^4 - (BC)t^3 + (C^2 - BD)t^2 + (2CD)t + D^2 = 0$$

Finding position

By substituting time into the formula we find the impact position D_p :

$$D_p = P_t + D_t$$

$$D_p = P_t + t\left(V_t + \frac{A_t t}{2}\right)$$

Finding velocity vector

To find the velocity vector you need to substitute time into the formula :

$$V_p = \frac{P_t}{t} + V_t - \frac{At}{2}$$